

## EXP 9: LINEAR AND POLYNOMIAL REGRESSION

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### AIM:

To compare the performance of Linear Regression and Polynomial Regression algorithms using Python on a given dataset.

### PROCEDURE

1. Import the required libraries and create a sample dataset.
2. Train the Linear Regression model and predict the output values.
3. Convert the input data into polynomial features.
4. Train the Polynomial Regression model and predict the output values.
5. Compare both models using the  $R^2$  score and Mean Squared Error (MSE).

### PROGRAM:

```
# Compare Linear Regression and Polynomial Regression
```

```
import numpy as np
```

```
from sklearn.linear_model import LinearRegression
```

```
from sklearn.preprocessing import PolynomialFeatures
```

```
from sklearn.metrics import r2_score, mean_squared_error
```

```
# Sample dataset
```

```
X = np.array([1,2,3,4,5,6,7,8,9,10]).reshape(-1,1)
```

```
y = np.array([1.5,3.8,8.2,15.5,24.5,35.8,49.2,64.5,82.0,101.5])
```

```
# Linear Regression
```

```
lr = LinearRegression()
```

```
lr.fit(X, y)
```

```
y_lr = lr.predict(X)
```

```
# Polynomial Regression (Degree = 2)
```

```
poly = PolynomialFeatures(degree=2)
```

```
X_poly = poly.fit_transform(X)
```

```
pr = LinearRegression()
```

```
pr.fit(X_poly, y)
```

```
y_pr = pr.predict(X_poly)
```

```
# Results
```

```
print("Linear Regression")
```

```
print("R2 Score:", r2_score(y, y_lr))
```

```
print("MSE:", mean_squared_error(y, y_lr))
```

```
print("\nPolynomial Regression")
```

```
print("R2 Score:", r2_score(y, y_pr))
```

```
print("MSE:", mean_squared_error(y, y_pr))
```

## OUTPUT:

```
Linear Regression  
R2 Score: 0.9447273748242438  
MSE: 60.108896969697  
  
Polynomial Regression  
R2 Score: 0.9999685238158671  
MSE: 0.03423030303030339
```

## RESULT:

The Logistic Regression algorithm was successfully implemented in Python. The model was trained and tested on the dataset, and its performance was evaluated using the confusion matrix and accuracy, achieving effective classification results.